

# SPATIO-TEMPORAL MODELING OF URBAN EXTREME RAINFALL EVENTS AT HIGH RESOLUTION

Rencontres Montpellier–Sherbrooke — June 4, 2026

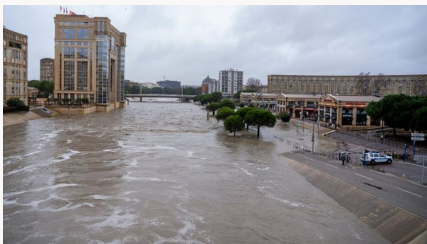
**Chloé SERRE-COMBE** <sup>1</sup>   Nicolas MEYER <sup>1</sup>   Thomas OPITZ <sup>2</sup>   Gwladys TOULEMONDE <sup>1</sup>

<sup>1</sup>IMAG, Université de Montpellier, LEMON Inria

<sup>2</sup>INRAE, BioSP, Avignon

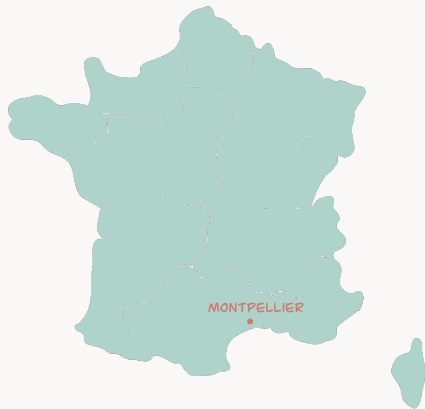


# GENERAL CONTEXT - MONTPELLIER, FRANCE



Floods in Montpellier, August 2015 and December 2025 (*Midi Libre*)

- Mediterranean events, localized rainfall
- Urban area, flood risks



# RAIN GAUGE NETWORK - OMSEV<sup>1</sup>

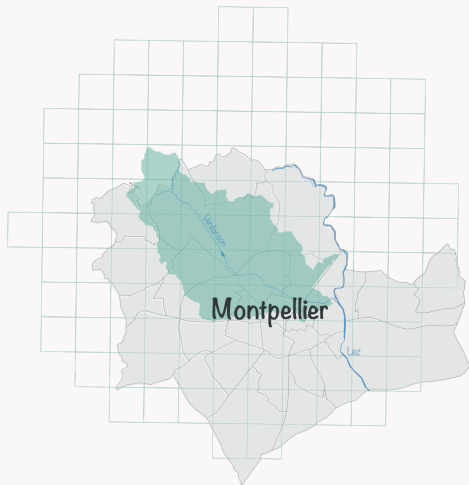


- **Study area:** Verdanson water catchment
- **Source:** Urban observatory of HydroScience Montpellier (HSM)<sup>2</sup>
- **Time period:** [Sept.2019, Jan.2025[
- **Temporal resolution:** 5 minutes
- **Spatial resolution:** 77 m to 2259 m

<sup>1</sup>Observ. Montpellierain et au Sud de l'Eau dans la Ville

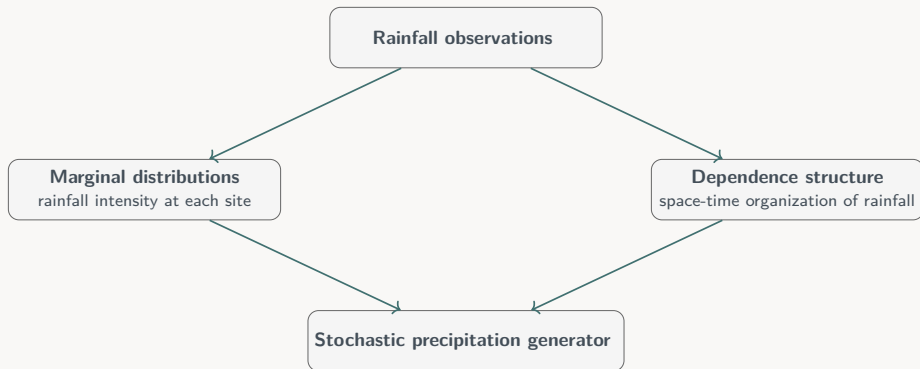
<sup>2</sup>FINAUD-GUYOT et al., 2023

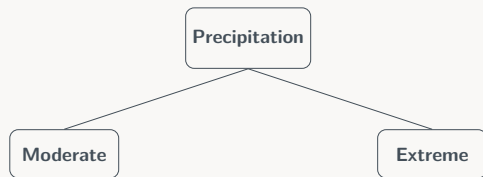
$$\mathcal{S} = \{20 \text{ rain gauges}\} \subset \mathbb{R}^2 \text{ and } \mathcal{T} \subset \mathbb{R}_+$$



- ▶ **Source:** COMEPHORE, Météo France
- ▶ **Time period:** [1997, 2025]
- ▶ **Temporal resolution:** Every hour
- ▶ **Spatial resolution:** 1 km<sup>2</sup>

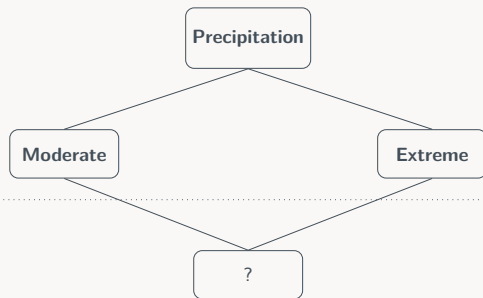
**Objectives:** reproduce statistical properties of rainfall for risk assessment.



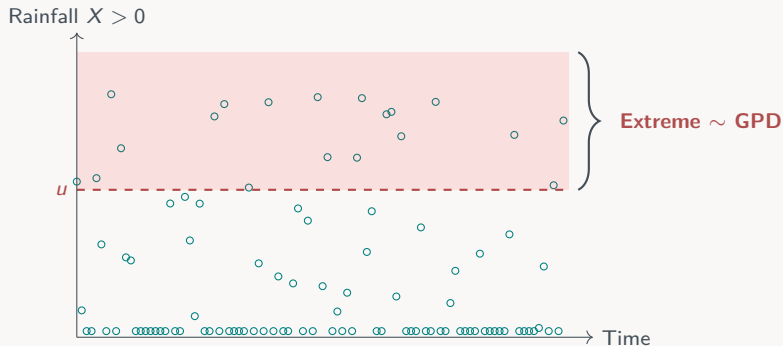


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Univariate



Univariate

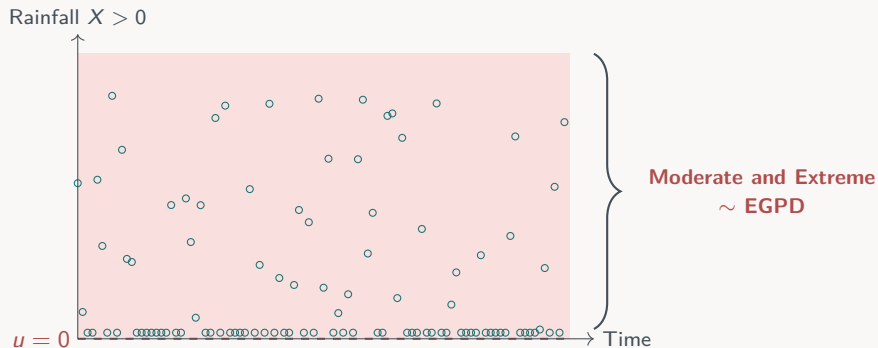


## ► Generalized Pareto Distribution (GPD)<sup>1</sup>

$$X \mid X > u \sim \underbrace{H_\xi}_{\text{GPD}(\xi, \sigma, u)} \text{ with } \xi \in \mathbb{R}, \sigma > 0.$$

$\xi$ : shape (tail behavior),  $\sigma$ : scale,  $u$ : threshold

<sup>1</sup>PICKANDS III, 1975



► **Extended Generalized Pareto Distribution (EGPD)<sup>2</sup>**

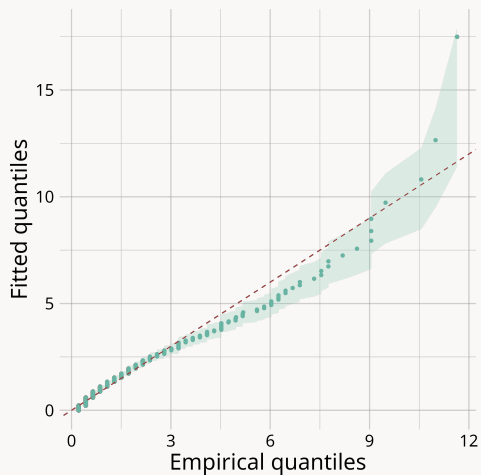
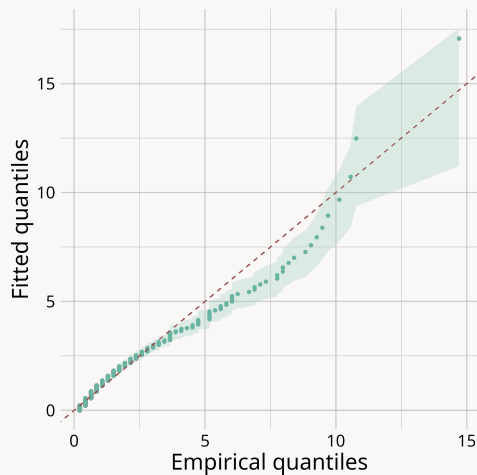
$$X \sim \underbrace{G(H_\xi)}_{\text{EGPD}(\xi, \sigma, \kappa)} \text{ with } G(x) = x^\kappa, \kappa > 0$$

<sup>2</sup>NAVEAU et al., 2016


$$\bar{\mu}(x) = \mu(x, (X))$$

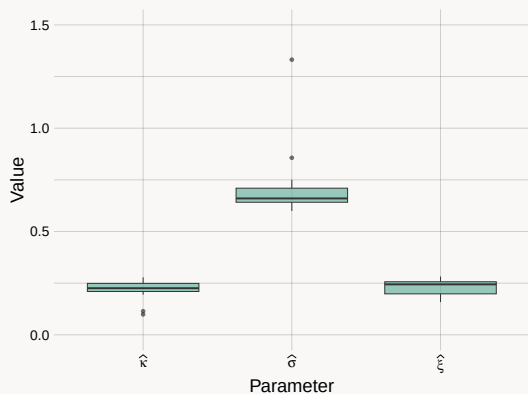
where  $G(x) = x^\kappa$ ,  $\kappa > 0$

- $\kappa$ : controls the bulk of the distribution
- Models **moderate and extreme** precipitation
- Avoids a threshold choice



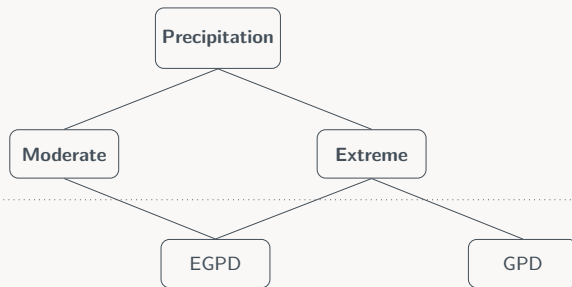
EGPD fitting for two rain gauges, CRBM (left) and UM (right) with left-censoring and 95% CI

# EGPD ESTIMATED PARAMETERS



Estimated parameters across rain gauges

- ▶  $\hat{\xi} \approx 0.25$ : extreme rainfall tail
- ▶  $\hat{\kappa} \approx 0.25$ : convective rainfall
- ▶  $\hat{\sigma} \approx 0.6$ : short-duration scale



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Univariate

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Dependence

**Rainfall random field:**  $\mathbf{X} = \{X_{s,t}, (s, t) \in \mathcal{S} \times \mathcal{T}\}$

Let  $\Lambda_{\mathcal{S}} \subset \mathbb{R}_+^2$  and  $\Lambda_{\mathcal{T}} \subset \mathbb{R}_+$  be sets of spatial and temporal lags respectively.

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## Variogram (MATHERON, 1963)

$$\gamma(\mathbf{h}, \tau) = \frac{1}{2} \text{Var}(X_{s,t} - X_{s+\mathbf{h}, t+\tau})$$

- Quantifies variability
- Higher  $\gamma(\mathbf{h}, \tau) \rightarrow$  weaker dependence

$$\mathbf{h} \in \Lambda_{\mathcal{S}}, \tau \in \Lambda_{\mathcal{T}}$$

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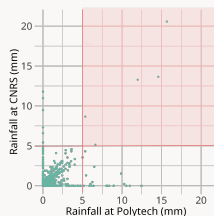
- Quantifies variability
- Higher  $\gamma(\mathbf{h}, \tau) \rightarrow$  weaker dependence

$\mathbf{h} \in \Lambda_{\mathcal{S}}, \tau \in \Lambda_{\mathcal{T}}, X_{s,t}^*$  uniform margins.

## Extremogram (DAVIS and MIKOSCH, 2009)

$$\chi(\mathbf{h}, \tau) = \lim_{q \rightarrow 1} \mathbb{P}(X_{s,t}^* > q \mid X_{s+\mathbf{h}, t+\tau}^* > q)$$

- Measures tail dependence
- Higher  $\chi(\mathbf{h}, \tau) \rightarrow$  stronger dependence

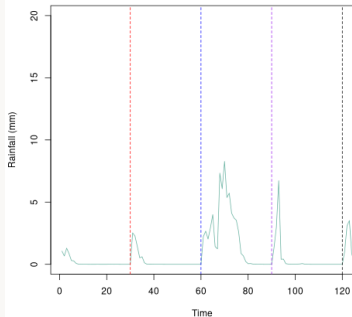


## Definition (DE FONDEVILLE and DAVISON, 2018)

For all  $s \in \mathcal{S}$  and  $t \in \mathcal{T}$ , a risk function  $r(\mathbf{X}) = X_{s_0, t_0}$ ,

$$u^{-1}X_{s,t} \mid X_{s_0, t_0} > u \xrightarrow{d} Y_{s,t} \quad \text{with} \quad Y_{s,t} = Re^{W_{s,t} - W_{s_0, t_0} - \gamma(s - s_0, t - t_0)},$$

where  $(s_0, t_0)$  is a space-time location,  $u$  is a high threshold,  $R \sim \text{Pareto}(1)$ ,  $W_{s,t}$  is a Gaussian process.



### ► $r$ -extremogram

$$\chi_r(\mathbf{h}, \tau) = \lim_{q \rightarrow 1} \mathbb{P}(X_{s_0 + \mathbf{h}, t_0 + \tau}^* > q \mid X_{s_0, t_0}^* > q)$$

### ► Brown-Resnick dependence structure

## Spatio-temporal extremogram with a Brown-Resnick dependence

Let  $\mathbf{h} \in \Lambda_{\mathcal{S}}$  and  $\tau \in \Lambda_{\mathcal{T}}$ . We have

$$\chi(\mathbf{h}, \tau) = 2 \left( 1 - \phi \left( \sqrt{\frac{1}{2} \gamma(\mathbf{h}, \tau)} \right) \right)$$

with  $\phi$  the std normal c.d.f. and  $\gamma$  the variogram of  $\mathbf{W}$ .

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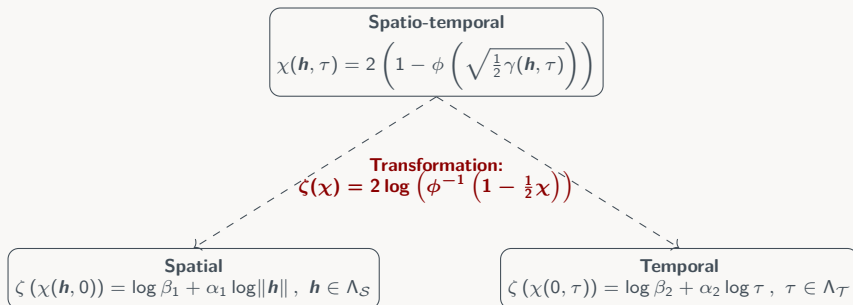
with  $\phi$  the std normal c.d.f. and  $\gamma$  the variogram of  $\mathbf{W}$ .

**Separable model:** Fractional Brownian motion with additive separability.

$$\frac{\gamma(\mathbf{h}, \tau)}{2} = \beta_1 \|\mathbf{h}\|^{\alpha_1} + \beta_2 |\tau|^{\alpha_2}$$

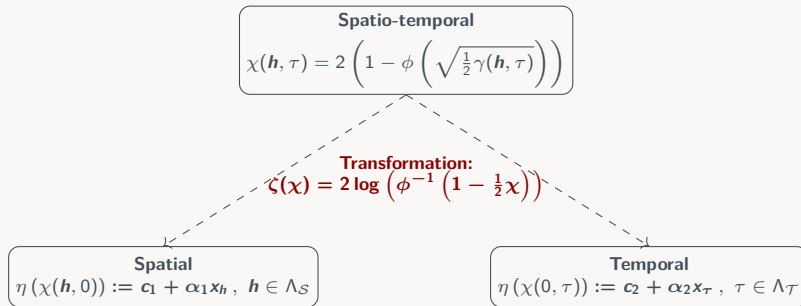
with  $0 < \alpha_1, \alpha_2 \leq 2$ ,  $\beta_1, \beta_2 > 0$  (BUHL et al., 2019).

**Case of additive separability:**  $\frac{\gamma(h, \tau)}{2} = \beta_1 \|h\|^{\alpha_1} + \beta_2 |\tau|^{\alpha_2}, \quad 0 < \alpha_1, \alpha_2 \leq 2, \quad \beta_1, \beta_2 > 0$



# ESTIMATION OF THE SEPARABLE VARIOGRAM PARAMETERS

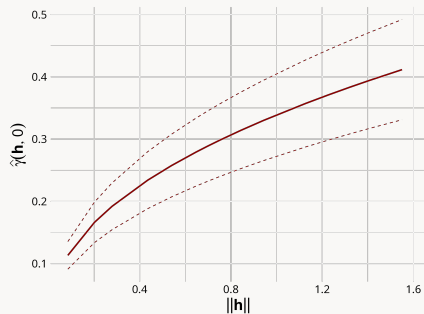
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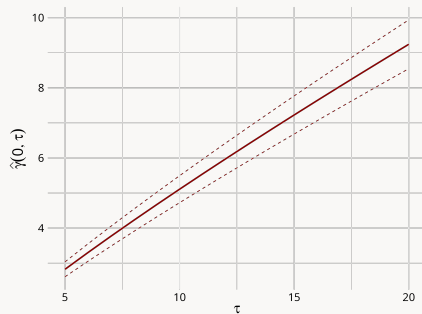
**Weighted Least Squares Estimation (WLSE)**

$$\begin{pmatrix} \hat{c}_i \\ \hat{\alpha}_i \end{pmatrix} = \operatorname{argmin}_{c_i, \alpha_i} \sum_x w_x (\zeta(\hat{x}) - (c_i + \alpha_i x))^2$$

# SEPARABLE DEPENDENCE ESTIMATION – WLSE RESULTS

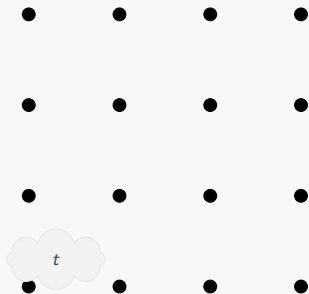


Spatial variogram  $\hat{\gamma}(\mathbf{h}, 0) = 2\hat{\beta}_1\|\mathbf{h}\|^{\hat{\alpha}_1}$

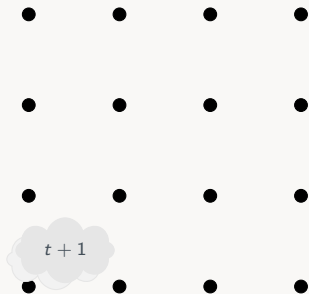


Temporal variogram  $\hat{\gamma}(\mathbf{0}, \tau) = 2\hat{\beta}_2|\tau|^{\hat{\alpha}_2}$

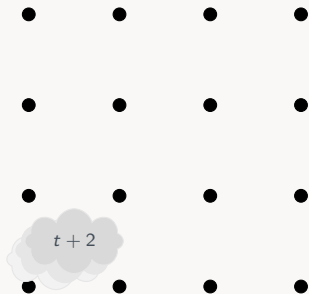
## Separable model



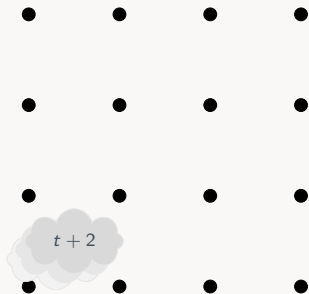
## Separable model



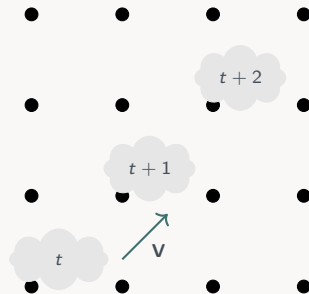
## Separable model



Separable model



Non-separable model



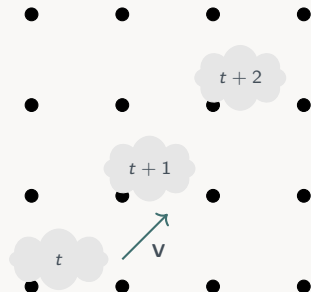
**Towards more realistic modeling:** introduce advection  $\mathbf{V}$  to relax separability

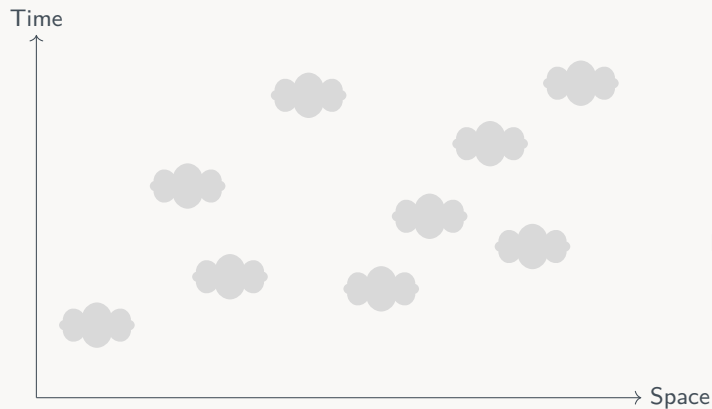
$$\gamma_L(\mathbf{h}, \tau) = \gamma(\mathbf{h} - \tau \mathbf{V}, \tau)$$

$$\Rightarrow \frac{1}{2} \gamma_L(\mathbf{h}, \tau) = \beta_1 \|\mathbf{h} - \tau \mathbf{V}\|^{\alpha_1} + \beta_2 |\tau|^{\alpha_2}$$

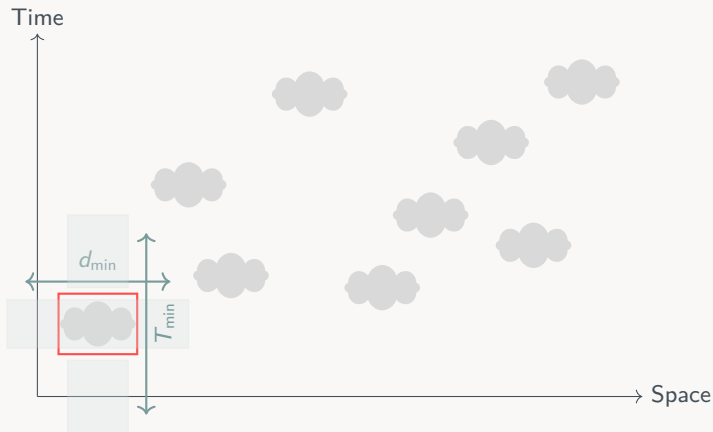
► Parameters:  $\Theta = (\beta_1, \beta_2, \alpha_1, \alpha_2, \mathbf{V})$

**Non-separable model**





Extreme episodes:  
Each episode is characterized  
by  $(s_0, t_0)$  for which  $X_{s_0, t_0} > u$



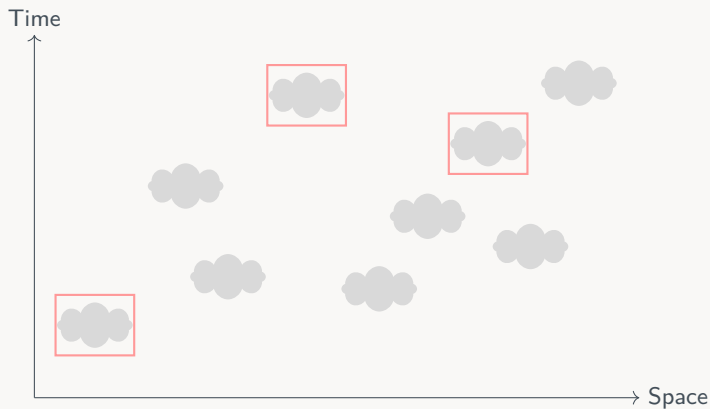
## Episode selection:

Only episodes separated by

- spatial distance  $\geq d_{\min}$
- OR

- temporal gap  $\geq T_{\min}$

$\Rightarrow$  reduces dependence  
between **selected episodes**  $\in \mathcal{E}$ .



## Episode selection:

Only episodes separated by

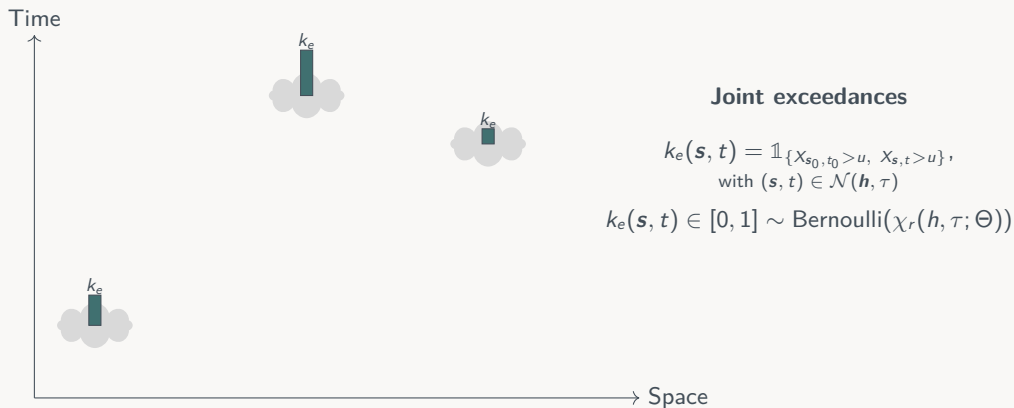
- spatial distance  $\geq d_{\min}$

OR

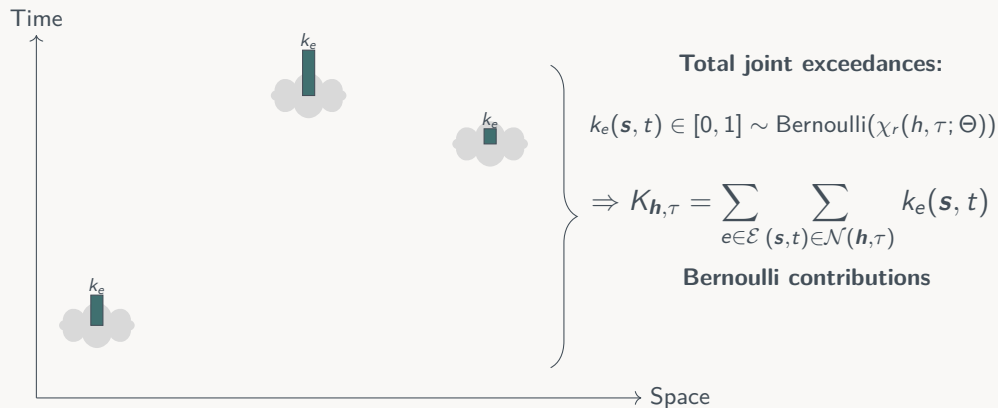
- temporal gap  $\geq T_{\min}$

$\Rightarrow$  reduces dependence  
between **selected episodes**  $\in \mathcal{E}$ .

# COUNT JOINT EXCEEDANCES PER EPISODE



with  $\mathcal{N}(\mathbf{h}, \tau) = \{(\mathbf{s}, t) \in \mathcal{S} \times \mathcal{T} \mid \|\mathbf{s} - \mathbf{s}_0\| = \|\mathbf{h}\|, |t - t_0| = \tau\}$



with  $\mathcal{N}(\mathbf{h}, \tau) = \{(\mathbf{s}, t) \in \mathcal{S} \times \mathcal{T} \mid \|\mathbf{s} - \mathbf{s}_0\| = \|\mathbf{h}\|, |t - t_0| = \tau\}$

**Bernoulli contributions (large  $u$ ):**

$$k_e(\mathbf{s}, t) = \mathbf{1}_{\{X_{s_0, t_0} > u, X_{\mathbf{s}, t} > u\}}, \quad (\mathbf{s}, t) \in \mathcal{N}(\mathbf{h}, \tau),$$

each treated as

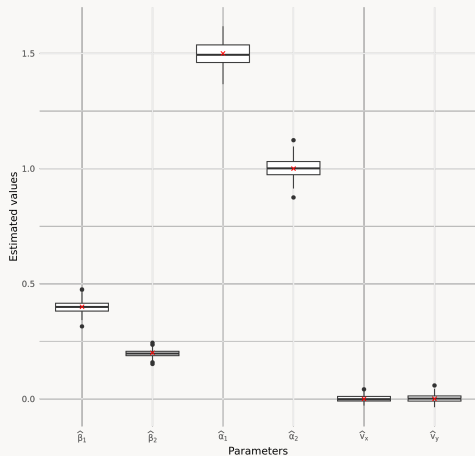
$$k_e(\mathbf{s}, t) \sim \text{Bernoulli}(\chi_r(\mathbf{h}, \tau; \Theta)).$$

### Composite log-likelihood

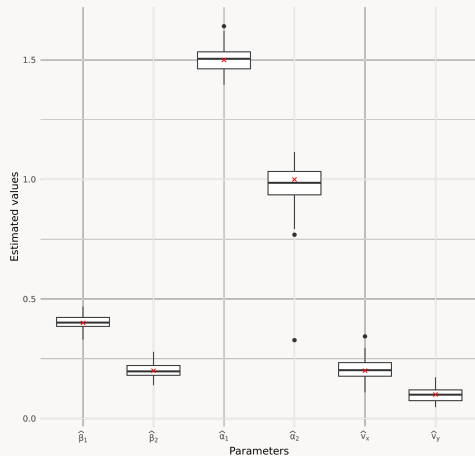
$$\ell_C(\Theta) \propto \sum_{e \in \mathcal{E}} \sum_{(\mathbf{h}, \tau) \in \Lambda_{\mathcal{S}} \times \Lambda_{\mathcal{T}}} \sum_{(\mathbf{s}, t) \in \mathcal{N}(\mathbf{h}, \tau)} k_e(\mathbf{s}, t) \log \chi_r(\mathbf{h}, \tau; \Theta) + (1 - k_e(\mathbf{s}, t)) \log(1 - \chi_r(\mathbf{h}, \tau; \Theta)).$$

- ▶ Optimization via maximization of  $\ell_C(\Theta)$
- ▶ Initial parameters: from WLSE of the separable model (i.e.,  $\mathbf{V} = \mathbf{0}$ )

# VALIDATION: CONSTANT ADVECTION



Without advection



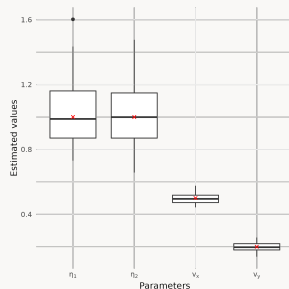
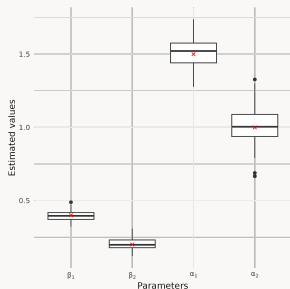
With advection

50 simulations, 500 replicates, 25 sites, 30 time steps

# EPISODE-WISE ADVECTION TO GLOBAL MODEL

$$\mathbf{V}^{\text{emp}} = \begin{bmatrix} \mathbf{V}^{(1)} \\ \mathbf{V}^{(2)} \\ \vdots \\ \mathbf{V}^{(N)} \end{bmatrix} \in \mathbb{R}^{N \times 2} \xrightarrow{\text{global model}} \mathbf{V}^{\text{final}} = \underbrace{\eta_1 \|\mathbf{V}^{\text{emp}}\|^{\eta_2} \frac{\mathbf{V}^{\text{emp}}}{\|\mathbf{V}^{\text{emp}}\|}}_{\text{transformation} \Rightarrow \eta \text{ to estimate}}, \eta_1 > 0, \eta_2 > 0$$

one component = one episode



50 simulations, 500 replicates, 25 sites, 30 time steps

## EPISODE-WISE ADVECTION TO GLOBAL MODEL

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one component = one episode

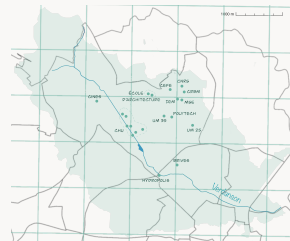
**Parameters:**  $\Theta = (\beta_1, \beta_2, \alpha_1, \alpha_2, \eta_1, \eta_2)$

### Challenge:

OMSEV data  $\Rightarrow$  limited information for advection

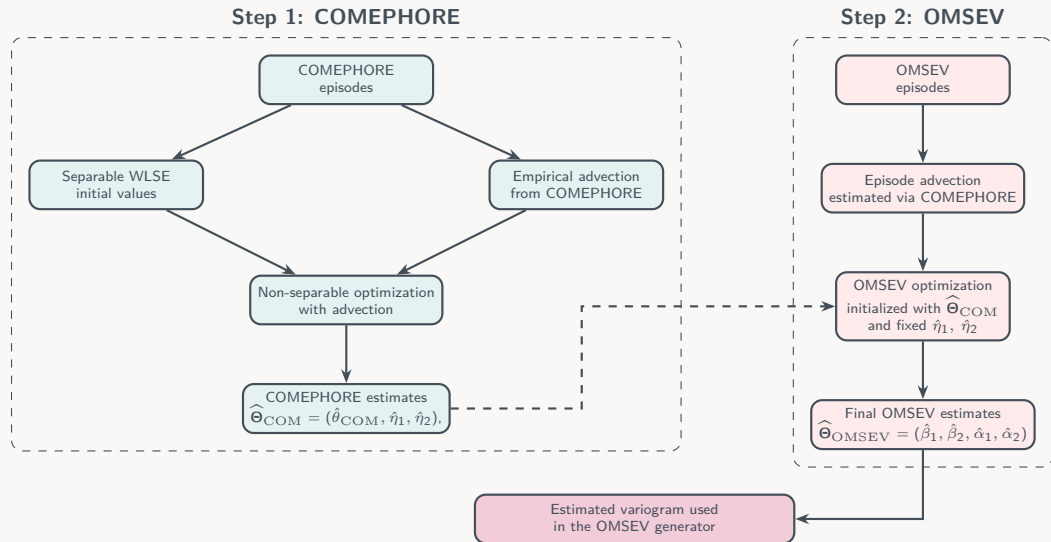
### Approach:

Fusion COMEPHORE-OMSEV → more reliable advection



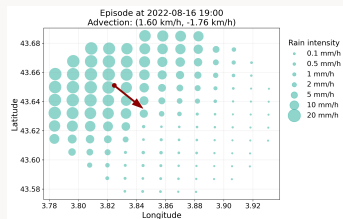
COMEPHORE pixels  
Météo France

# ESTIMATION STRATEGY FOR THE EXTREMAL DEPENDENCE STRUCTURE

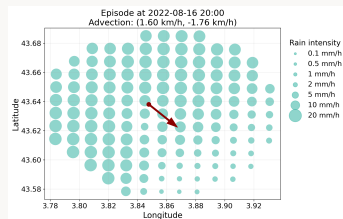


# ADVECTION ESTIMATION ON REAL DATA

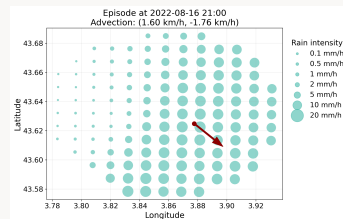
For one episode, we have:



$t_0 - 1$



$t_0$

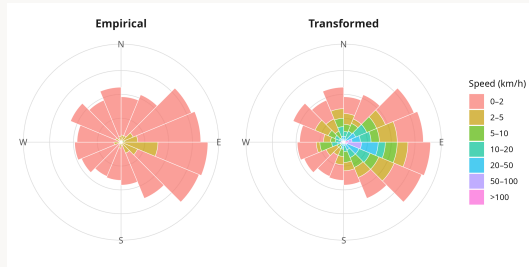


$t_0 + 1$

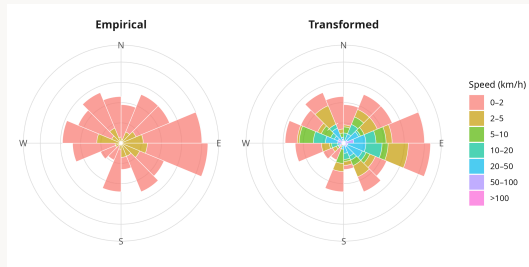
$\mathbf{V}^{\text{emp}}$  is estimated from **rain storm barycenter displacement** within a time window.

Optimization on COMEPHORE episodes  $\Rightarrow \hat{\eta}_1 = 3.896$  and  $\hat{\eta}_2 = 2.221$ .

# ADVECTION ESTIMATION ON REAL DATA

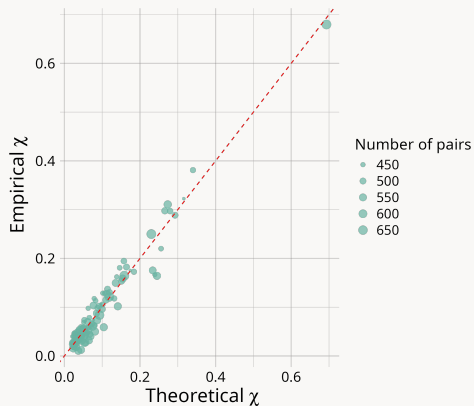


COMEPHORE



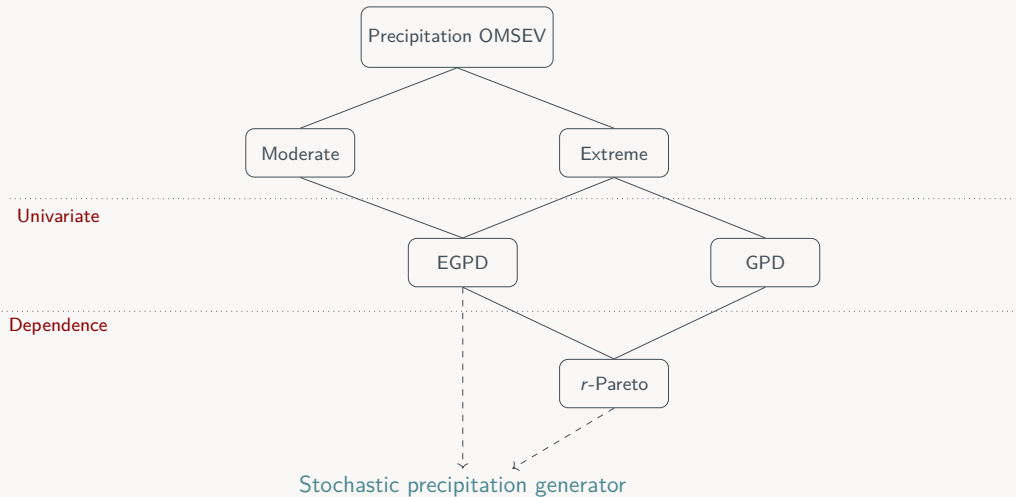
OMSEV

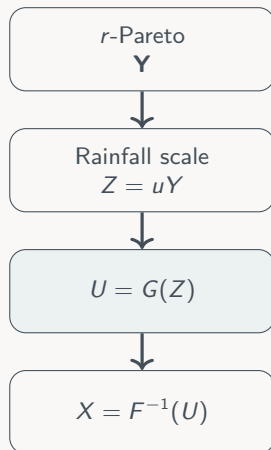
Parameter estimates over  $|\mathcal{E}| = 333$  episodes  
 $\hat{\eta}_1 = 3.896$  and  $\hat{\eta}_2 = 2.221$  fixed (COMEPHORE)



|                  | m/5 min              |
|------------------|----------------------|
| $\hat{\beta}_1$  | 0.231 [0.047, 1.060] |
| $\hat{\beta}_2$  | 0.807 [0.368, 1.655] |
| $\hat{\alpha}_1$ | 0.250 [0.064, 0.436] |
| $\hat{\alpha}_2$ | 0.666 [0.469, 0.863] |

# STOCHASTIC SIMULATION OF PRECIPITATION EVENTS





**Marginals: zero and EGPD**

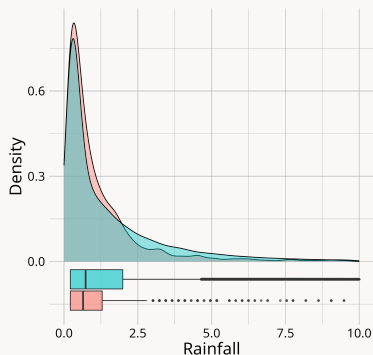
$$F(x) = \begin{cases} 0, & x < 0, \\ p_0, & x = 0, \\ p_0 + (1 - p_0)F_{\text{EGPD}}(x), & x > 0. \end{cases}$$

**Role of G:**

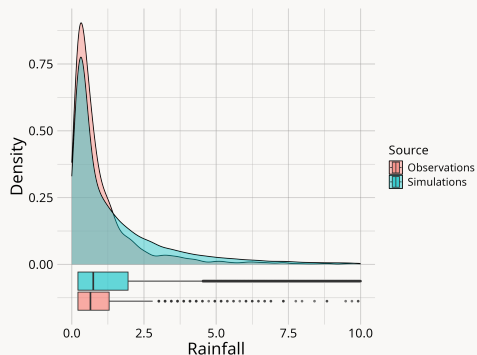
- maps to unit interval
- introduces mass at zero
- preserves Pareto tail

# MARGINAL VALIDATION OF SIMULATED PRECIPITATION

100 simulations by episode



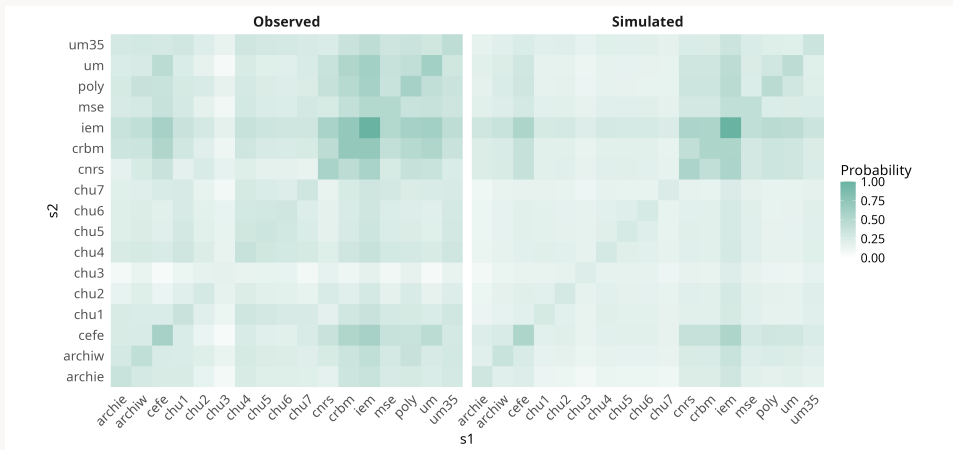
UM density



IEM density

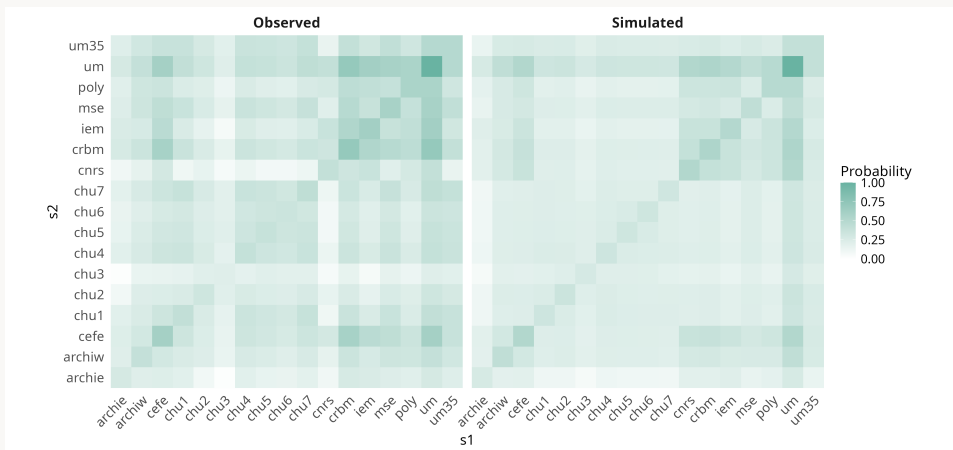
Comparison of marginal distributions (above-zero rainfall)

100 simulations by episode



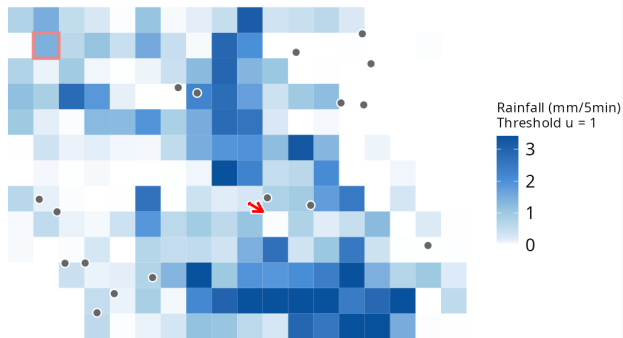
$$\mathbb{P}(X_{s_1} > u, X_{s_2} > u | X_{\text{IEM}} > u)$$

100 simulations by episode



$$\mathbb{P}(X_{s_1} > u, X_{s_2} > u | X_{UM} > u)$$

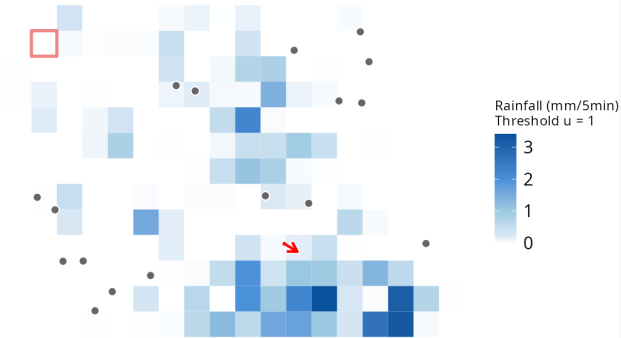
$t = 1$  | advection per 5 min = (50, -30) m



Over the OMSEV network area

- Spatial resolution: 100 m  $\times$  100 m
- Temporal resolution: 5 min
- Stochastic rainfall simulation

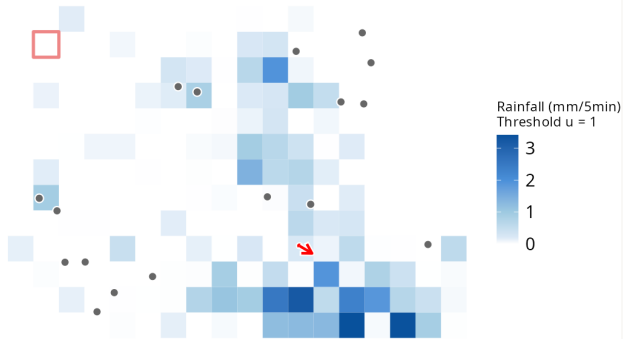
$t = 2$  | advection per 5 min = (50, -30) m



**Over the OMSEV network area**

- Spatial resolution: 100 m  $\times$  100 m
- Temporal resolution: 5 min
- Stochastic rainfall simulation

$t = 3$  | advection per 5 min = (50, -30) m



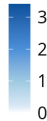
## Over the OMSEV network area

- Spatial resolution: 100 m  $\times$  100 m
- Temporal resolution: 5 min
- Stochastic rainfall simulation

$t = 4$  | advection per 5 min = (50, -30) m

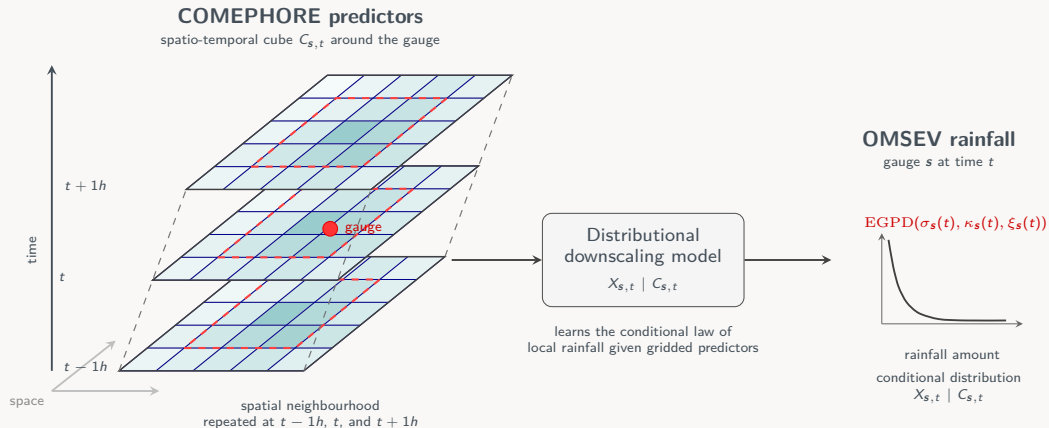


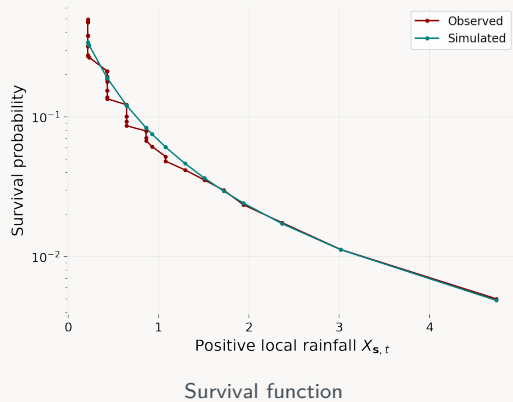
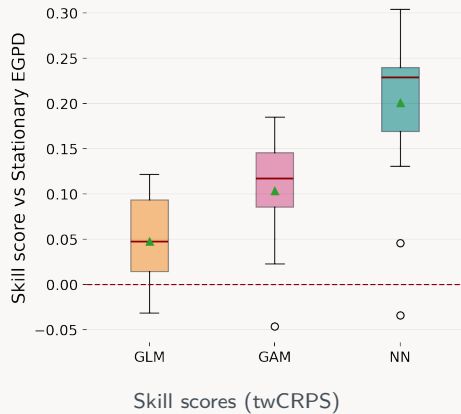
Rainfall (mm/5min)  
Threshold  $u = 1$



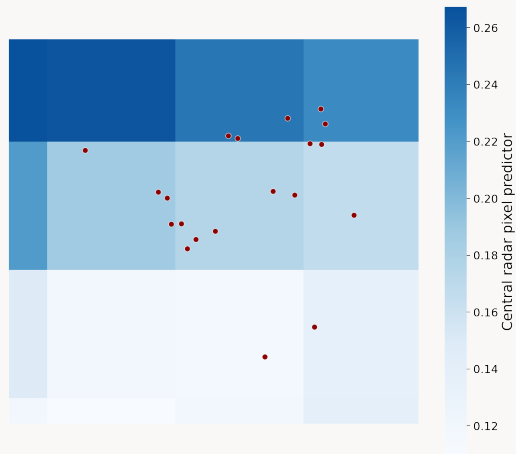
## Over the OMSEV network area

- Spatial resolution: 100 m  $\times$  100 m
- Temporal resolution: 5 min
- Stochastic rainfall simulation

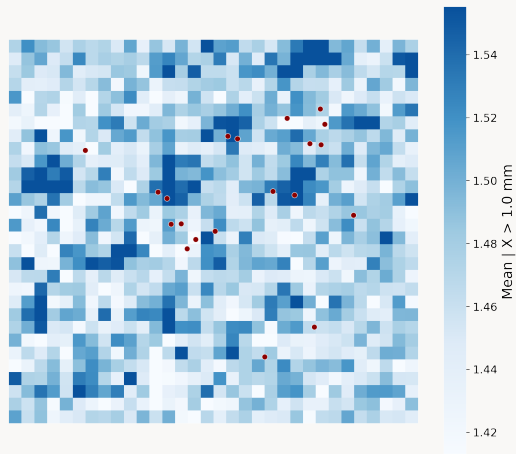




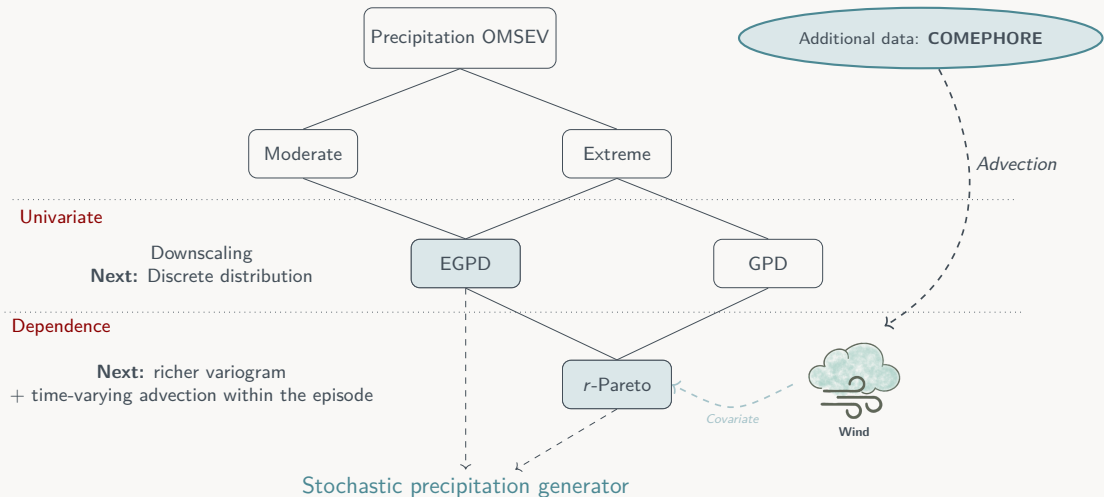
## Monthly simulations - September 2022



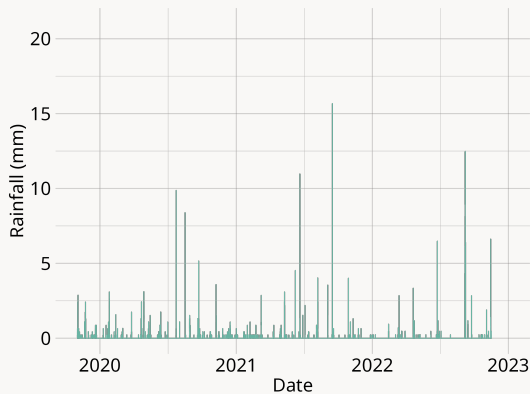
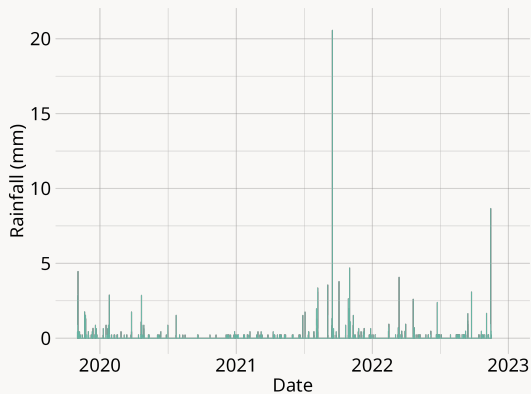
Central predictor rainfall



Mean excess



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- DAVIS, R. A., & MIKOSCH, T. (2009). The extremogram: A correlogram for extreme events.
- DE FONDEVILLE, R., & DAVISON, A. C. (2018). High-dimensional peaks-over-threshold inference. *Biometrika*, 105(3), 575–592.
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- MATHERON, G. (1963). Principles of geostatistics. *Economic geology*, 58(8), 1246–1266.
- NAVEAU, P., HUSER, R., RIBEREAU, P., & HANNART, A. (2016). Modeling jointly low, moderate, and heavy rainfall intensities without a threshold selection. *Water Resources Research*.
- PICKANDS III, J. (1975). Statistical inference using extreme order statistics. *the Annals of Statistics*, 119–131.



Rainfall amounts on CNRS and Polytech rain gauges

## Generalized Pareto Distribution



## Extended GPD<sup>1</sup>

$$\overline{H}_\xi \left( \frac{x-u}{\sigma} \right) = \begin{cases} (1 + \xi \frac{x-u}{\sigma})_+^{-1/\xi} & \text{if } \xi \neq 0, \\ e^{-\frac{x-u}{\sigma}} & \text{if } \xi = 0, \end{cases}$$

where  $a_+ = \max(a, 0)$ ,  $\sigma > 0$ ,  $x - u > 0$

- Models extreme precipitation
- Depends on a threshold choice

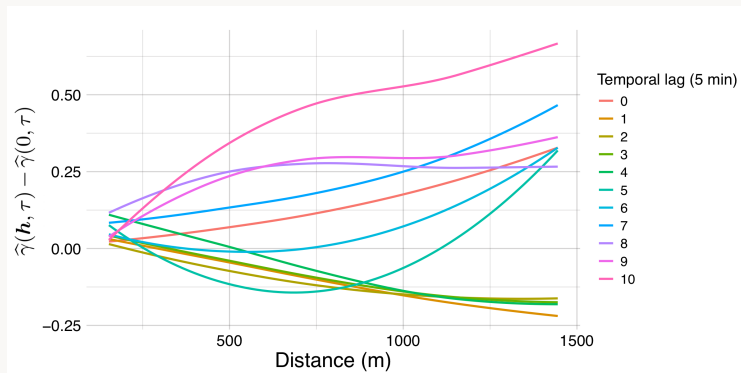
$$F(x) = G \left( H_\xi \left( \frac{x}{\sigma} \right) \right),$$

where  $G(x) = x^\kappa$ ,  $\kappa > 0$

- Models **moderate and extreme** precipitation
- Avoids a threshold choice

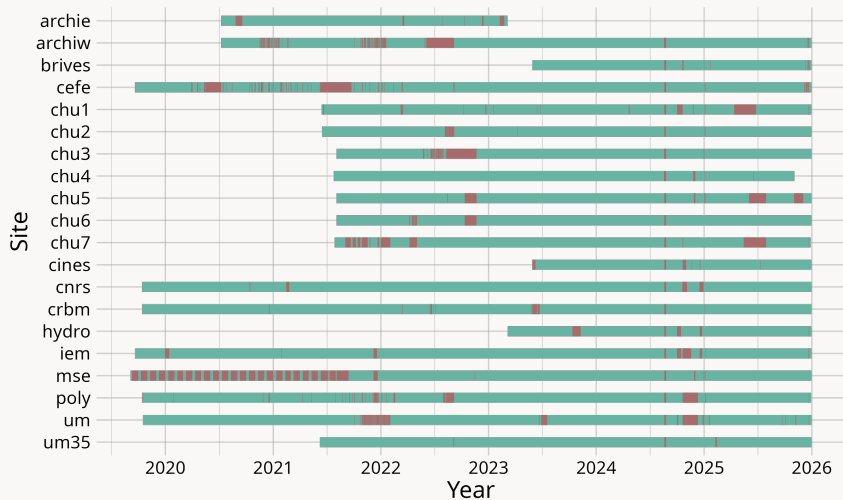
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<sup>1</sup>NAVEAU et al., 2016



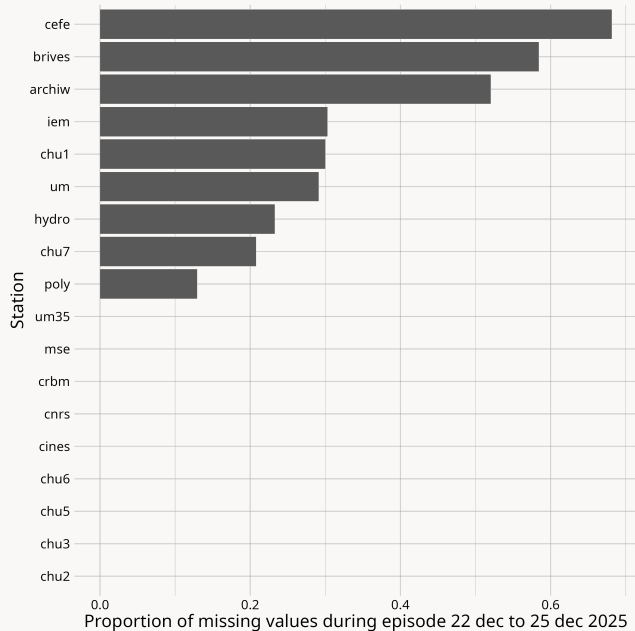
Empirical difference between the spatio-temporal variogram  $\gamma(\mathbf{h}, \tau)$  and the only temporal variogram on OMSEV data

# DATA ISSUE: MISSING VALUES



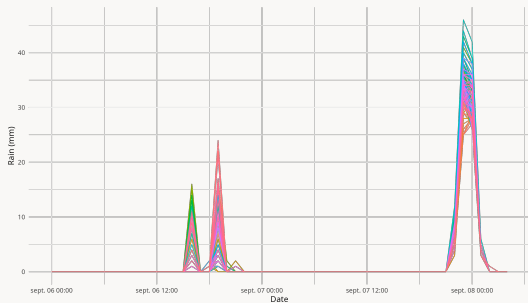
Missing values according to site activation periods (OMSEV dataset)

## DATA ISSUE: MISSING VALUES

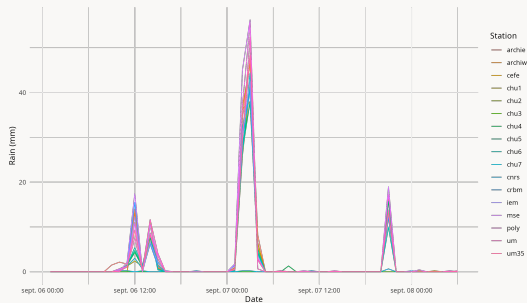


# DATA COMPARISON: RADAR VS RAIN GAUGES

## COMEPHORE (radar)



## OMSEV (gauges)



Mismatch between radar and gauge data